

DRAFT

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Question #: 1

A 13-year-old boy is brought to the physician by his mother because of a 2-day history of sore throat, runny nose and cough. His blood pressure is 90/70 mmHg, pulse is 90/min, respirations are 20/min and temperature is 38°C (100°F). Physical examination shows normal lung sounds. A diagnosis of an upper respiratory tract infection is made. Which of the following germ cell layers is the epithelial lining of the infected system most likely derived from?

- A. Endoderm
- B. Ectoderm
- C. Paraxial mesoderm
- D. Intermediate mesoderm
- E. Lateral mesoderm

Question #: 2

A 56-year-old man is hospitalized following a car crash. Parenteral nutrition is started and dietary essential fatty acids of the omega-3 family are included in the supplements. Which of the following fatty acids most likely belongs to this group and is supplemented in this patient?

- A. Palmitoleic acid
 - B. Oleic acid
 - C. Linoleic acid
 - D. Alpha-linolenic acid
 - E. Arachidonic acid
 - F. Palmitic acid
 - G. Stearic acid
 - H. Docosahexaenoic acid
 - I. Eicosapentaenoic acid
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Question #: 3

A researcher is analyzing the cell membrane of neuronal tissues and identifies that it has a high content of glycolipids with N-acetyl neuraminic acid (NANA) on the outer leaflet of the plasma membrane. Which of the following compounds most likely belongs to this class of lipids?

- A. Phosphatidylethanolamine
- B. Phosphatidylinositol
- C. Sphingomyelin
- D. Ganglioside
- E. Triacylglycerol
- F. Globosides
- G. Cerebrosides
- H. Cardiolipin

Question #: 4

A 49-year-old man comes to the physician because of a 2-month history of skin lesions. Laboratory and clinical studies show stage IVB - cutaneous T-cell lymphoma. Vorinostat (Zolinza) which is an anticancer drug is prescribed to the patient. Vorinostat is a member of a large class of drugs that act as histone deacetylase inhibitors (HDAC inhibitors). Which of the following is the most likely mechanism of action of this drug?

- A. Causes histones to become positively charged
- B. Induces the formation of euchromatin
- C. Decreases gene transcription
- D. Increases the affinity of DNA to the nucleosome
- E. Inhibits DNA methylation

Question #: 5

A 32-year-old woman comes to the physician because of a 4-day history of watery diarrhea following a week-long hiking trip. Physical examination shows diffuse abdominal tenderness and no evidence of peritoneal irritation. Stool examination for ova and parasites shows the presence of *Giardia intestinalis* cysts. This pathogen destroys the brush border of the intestines and the intestinal villi. Which of the following glucose transporters is most likely to be directly affected as a result of this infection?

- A. GLUT-4
- B. SGLT
- C. GLUT-2
- D. GLUT-1
- E. GLUT-3

Question #: 6

Nasal epithelial cells are isolated from human nasal brushings and bathed in an extracellular solution containing: Na = 140 mmol, K = 5 mmol, Cl 105 mmol. Assume normal intracellular electrolyte concentrations of Na = 14 mmol, K = 140 mmol and Cl = 40 mmol. The membrane potential of one of the epithelial cells is measured at -60 mV. Assume that the constant $RT/F = 60$ mV. What is the driving force and direction of flux for sodium across the cell membrane?

- A. 120 mV influx
- B. 60 mV influx
- C. 30 mV influx
- D. 30 mV efflux
- E. 60 mV efflux
- F. 120 mV efflux

Question #: 7

A research team is developing a drug that targets a secondary active antiporter. Which of the following processes is most likely an example of secondary active anti-transport?

- A. Glucose uptake into skeletal muscle
- B. Transport of glucose from the intestinal lumen into an intestinal epithelial cell
- C. Movement of the Na^+ ions into a nerve cell during the upstroke of an action potential
- D. Na^+ -dependent transport of Ca^{2+} from the cytosol to the extracellular fluid
- E. Ouabain-sensitive transport of Na^+ ions from the cytosol to the extracellular fluid

Question #: 8

Movement of Na^+ ions into a nerve cell during the upstroke of an action potential is being studied in a model system. Which of the following transport processes would most likely be

observed during this study?

- A. Primary active transport
- B. Secondary co-transport
- C. Secondary counter-transport
- D. Facilitated diffusion
- E. Passive diffusion through the cell membrane

Question #: 9

A 25-year-old woman mountaineer comes to see her physician, explaining that she plans to climb Mount Everest. Her vital signs are within normal limits and physical examination shows no abnormalities. In preparation for the climb, she flies directly from sea level to base camp at an altitude of 5000 m. On reaching base camp, her ventilation rate increases as a result of the low PO₂ activating her peripheral chemoreceptors. The increased ventilation then reduces her blood CO₂ levels, which then trigger the central chemoreceptors to reduce ventilation rate. Safeguarding arterial oxygen is deemed more important than removing CO₂, and her ventilation rate continued to be marginally elevated. What name is given to the homeostatic process outlined in this example?

- A. Hierarchy
- B. Redundancy
- C. Feed forward regulation
- D. Positive feedback
- E. Competitive antagonism

Question #: 10

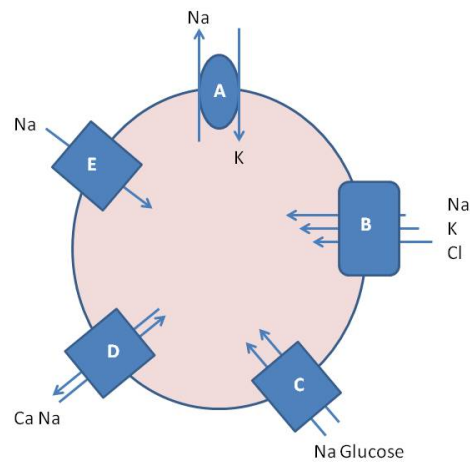
A group of researchers is investigating primary active transport mechanisms in a cultured human cell line. The attached diagram depicts the active and expressed transport proteins in these cells. Which labelled transport process would be most immediately affected by uncoupling mitochondria and limiting ATP production?

- A. A
- B. B
- C. C

D. D

E. E

Attachment:



41132.jpg

Question #: 11

A 32-year-old woman in the third trimester of her pregnancy comes to the emergency room because of a 1-hour history of increasingly frequent and forceful uterine contractions. An increase in uterine muscle tension leads to the release of oxytocin from the pituitary gland. Oxytocin then stimulates a further increase in uterine muscle contractions and more tension develops. Which of the following homeostatic mechanisms best describes this phenomenon?

- A. Negative Feedback
- B. Redundancy
- C. Feed-Forward Regulation
- D. Hierarchy
- E. Positive Feedback

Question #: 12

A 24-year-old man is brought to the emergency room unconscious. The ambulance driver says on entering the ambulance, the patient was conscious, struggling to breathe, hypersalivating, sweating and demonstrating uncoordinated behavior. During the 15-minute journey to hospital he lost consciousness and started to seize. His wife says that in the restaurant he complained of

numbness and tingling around his lips and extremities after consuming pufferfish. Pufferfish contains tetrodotoxin, a voltage-gated sodium channel blocker. His pulse is 95/min, blood pressure is 110/90 mm Hg and respirations are 5/min. He is immediately placed on oxygen after gastric lavage and given an IV bolus of alpha adrenoceptor agonist to increase his blood pressure. Which of the following pathophysiological findings is most likely seen in this patient's action potentials?

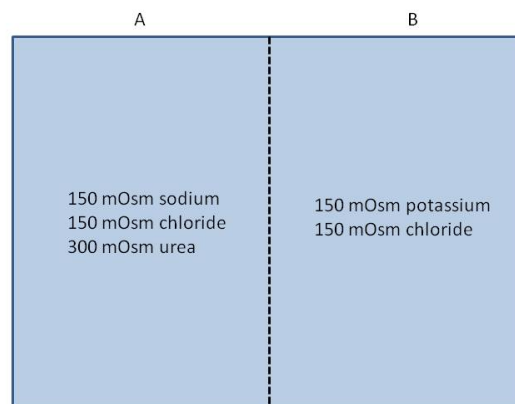
- A. Complete absence of action potential formation
- B. Slowed rate of depolarization
- C. Delayed repolarization
- D. Absence of repolarization
- E. Slowed rate of repolarization

Question #: 13

Membrane properties are being studied in a setup where a semi-permeable membrane (dashed line) separates 2 compartments, as shown in the attached image. Which of the following best describes the tonicity and osmolarity of solution A compared to solution B, as illustrated in the attached image?

- A. Isotonic and hyperosmotic
- B. Hypertonic and hyperosmotic
- C. Hypotonic and hyperosmotic
- D. Isotonic and iso-osmotic
- E. Hypertonic and hypo-osmotic

Attachment:



Reflection Coefficients
Sodium = 1
Potassium = 1
Chloride = 0
Urea = 0

Question #: 14

A healthy 21-year-old man is enrolled in an exercise physiology trial examining muscle fatigue. Physical examination is normal and shows him to be in excellent condition. Applied kinesiology and dynamic muscle function tests show normal nerve conduction and normal muscle strength at the beginning of the experiment. Which of the following mechanisms best explains the afterhyperpolarization of the motor action potentials observed in this man?

- A. Increased sodium/potassium pump activity
- B. Increased fractional sodium conductance
- C. Increased potassium conductance
- D. Increased fractional potassium conductance
- E. Increased sodium conductance

Question #: 15

A 23 year-old man comes to the physician 2 hours after a marathon run on a hot day because he is feeling very faint and dizzy. Isotonic fluid replacement is considered to be the best course of treatment. Assuming standard cellular concentrations and osmolarity (300 mOsm), which of the following solutions would most likely be isotonic?

- A. 150 mM NaCl + 20 mM Urea
- B. 300 mM NaCl
- C. 150 mM NaCl + 20 mM dextrose
- D. 300 mM NaCl + 20 mM Urea
- E. 300 mM NaCl + 20 mM dextrose

Question #: 16

A 15-year-old boy is brought to the physician because of a 2-day history of muscle weakness and fatigue, especially after exercise. He says his limbs feel 'like rubber'. Laboratory studies of blood electrolyte levels are shown in the table below. Using the values in table, which of the following values best approximates the electrochemical equilibrium potential of potassium in this patient?

Ion	extracellular (mM)	intracellular (mM)
Na^+	140	14
K^+	2.4	120
Cl^-	104	4
Ca^{2+}	2.1	10^{-6}

- A. -120 mV
- B. -102 mV
- C. -88 mV
- D. -71 mV
- E. -54 mV

Question #: 17

A 16-year-old boy is brought to the emergency room after being unable to walk and struggling to stand without help after competing at an athletics event. He says he feels extremely weak. He has a family history of primary hypokalemic periodic paralysis. Laboratory studies show that his plasma potassium level is 2.1 mmol/L (normal range 3.5 –5 mmol/L). He is treated with a K^+ supplement in a saline IV infusion. Which of the following is the most likely primary cause of his weakness?

- A. Depolarized rest membrane potential of excitable cells
- B. Hyperpolarized resting membrane potential of excitable cells
- C. Increased refractory period of voltage gated sodium channels
- D. Depolarizing blockade of voltage gated sodium channels
- E. Increased opening of delayed rectifier potassium channels

Question #: 18

A 22-year-old man comes to the physician because of a 2-day history of diarrhea. His blood pressure is 128/86 mm Hg, pulse is 64/min and respirations are 16/min. Increasing ventilation of the lungs will result in lower plasma CO_2 concentrations, making the blood more alkaline and suppressing breathing. Conversely, decreasing ventilation will result in higher plasma CO_2 levels, causing increased

acidity and stimulating breathing. This homeostatic regulation is best described by which of the following mechanisms?

- A. Negative feedback
- B. Positive feedback
- C. Feed-forward regulation
- D. Redundancy
- E. Change of homeostatic setpoint

Question #: 19

A 56-year-old man comes to the emergency department because of a 2-week history of generalized weakness, dizziness, and fatigue and a 1-day history of palpitations and fainting. His blood pressure is 90/60 mmHg and pulse is 140/min and irregular. Electrocardiogram shows atrial fibrillation. The physician terminates the atrial fibrillation in the emergency department, and the patient is discharged on digoxin. This medication is known to inhibit the sodium-potassium pump found in the cell's plasma membrane. Which of the following constituents of the cell's membrane is most likely affected in this case?

- A. Glycolipids
- B. Cholesterol
- C. Integral proteins
- D. Phospholipid
- E. Peripheral proteins

Question #: 20

A 40-year-old woman comes to the physician because of a 3-month history of a swelling on the medial aspect of her arm. Physical examination shows a non-tender, mobile mass in the same area. The physician suspects a lipoma, a benign tumor of adipocytes. A biopsy is obtained for histopathological examination. Which of the following can be used as a fixative and stain to preserve and visualize the lipid in these cells for microscopic study?

- A. Hematoxylin
- B. Eosin

- C. Silver
- D. Periodic acid Schiff
- E. Osmium tetroxide

Question #: 21

A 63-year-old woman comes to the physician for a routine colonoscopy. During the procedure a mass is observed in the ascending colon. A biopsy is taken and a karyotype shows aneuploidy within the cells taken from the mass. The epithelium of the affected structure is usually a single cell layer thick with cells that are taller than they are wide. Which of the following best characterizes this epithelium?

- A. Simple columnar
- B. Transitional
- C. Pseudostratified ciliated
- D. Stratified cuboidal
- E. Stratified squamous

Question #: 22

A 32-year-old woman comes to the physician because of a 1-week history of fluid-filled blisters on her upper limbs. Electron microscopy of **biopsy of these areas**, show the separation of epithelial cells from the basement membrane. Which of the following cell adhesion molecules is a component of the junctional complexes found in this region?

- A. Selectin
- B. Cadherin
- C. Integrin
- D. Desmoglein
- E. Claudin

Question #: 23

A transmission electron microscopic study of endothelial cells of the lung vasculature involved in gas exchange is shown in the attached image. Which of the following mechanisms best describes the formation of these small, smooth vesicles (arrows) in the cytoplasm of these cells?

- A. Autophagy
- B. Exocytosis
- C. Phagocytosis
- D. Pinocytosis
- E. Receptor-mediated endocytosis

Attachment:



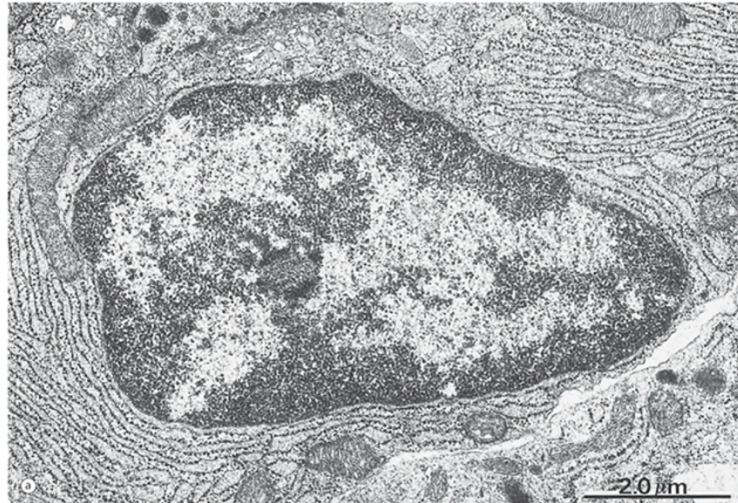
Question 7.jpg

Question #: 24

A biomedical scientist is investigating the effect of a tumor suppressant on an aggressive tumor. The drug is found to inhibit the function of cells which have features similar to the cell in the attached photomicrograph. Which of the following functions will the drug most likely be inhibiting?

- A. Steroid hormone production
- B. Phagocytosis
- C. Enzyme secretion
- D. Contraction
- E. Lipid storage

Attachment:



secretory cell 1.png

Question #: 25

A biotech company is conducting research on a new compound that functions as a chemotactic agent. Cell migration experiments show that neutrophils form lamellipodia and move towards the new compound in cell culture. Which of the following is most likely responsible for the formation of this cellular protrusion?

- A. Actin polymerization
- B. Intermediate filament polymerization
- C. Kinesin
- D. Microtubule polymerization
- E. Myosin

Question #: 26

A 21-year-old woman comes to the physician because of a 6-month history of severe acne on her face. Physical examination shows some pimples are acutely inflamed and a few have some pus inside, which can be easily expressed out. The sebaceous glands involved in this condition have a secretory process

wherein the entire cell becomes a part of the secretory product. Which of the following best describes this mode of secretion?

- A. Merocrine
- B. Holocrine
- C. Autocrine
- D. Paracrine
- E. Apocrine

Question #: 27

An 8-year-old boy is brought to the physician by his mother because of a 3-day history of fever and a cough productive of greenish sputum. He is diagnosed with pneumonia (microbial infection of the respiratory passages). This is his 3rd episode of pneumonia in the last 6 months. Further testing shows that he has a dysfunction of the apical specialization cilia. Which of the following best characterizes the function of the affected structure?

- A. Increases the surface area for absorption
- B. Barrier against infections
- C. Compartmentalization of mitochondria
- D. Movement of trapped particles along surfaces
- E. Adhesion to neighboring cells
- F. Secretion of mucus

Question #: 28

A 32-year-old woman comes to the physician for a follow up examination. She received a stab wound in her abdomen 3 weeks ago. Physical examination today shows that the sides of the wound are progressing well toward each other. The physician notes that this action is contributed to by the influence of a certain junctional complex that links the cell's actin cytoskeleton to the extracellular matrix. Which of the following transmembrane protein is a component of this junctional complex?

- A. Selectin

- B. Cadherin
- C. Integrin
- D. Desmocollins
- E. Claudin

Question #: 29

A pharmaceutical company discovers a new compound that inhibits the release of a secreted product that is continuously delivered to the plasma membrane for export. Which of the following processes is most likely being inhibited by this compound?

- A. Phagocytosis
- B. Pinocytosis
- C. Exocytosis, regulated pathway
- D. Exocytosis, constitutive pathway
- E. Receptor-mediated endocytosis

Question #: 30

A 26-year-old man is brought to the emergency department because of a 30-minute history of shivering following the use of an illegal drug. His blood pressure is 90/40 mmHg, pulse is 123/min, and respirations are 21/min. Physical examination shows a cold, sweaty, and almost unconscious young man. Treatment of this patient requires knowledge of the presence of cytochrome P450 within liver cells (hepatocytes). In which of the following organelles within the hepatocyte is this enzyme most likely located?

- A. Lysosomes
- B. Peroxisomes
- C. Smooth endoplasmic reticulum
- D. Proteasomes
- E. Golgi apparatus